

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Develop a Dynamic Webpage Using DOM Manipulation and JavaScript

2. AIM

To develop a dynamic webpage that updates its content using JavaScript and DOM manipulation, without reloading the page.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the Document Object Model (DOM).
- To access and modify HTML elements using JavaScript.
- To handle events such as button clicks.
- To dynamically update webpage content and add new elements without reloading the page.

4. THEORY

The Document Object Model (DOM) represents an HTML document as a tree of objects that JavaScript can access and manipulate. Each HTML element becomes a node that can be read, changed, added or removed using JavaScript.

Methods such as `document.getElementById()` and `document.createElement()`, along with properties like `innerHTML` and `style`, allow JavaScript to dynamically read and update the content, structure and appearance of a webpage.

Event handling, using attributes like `onclick` or the `addEventListener()` method, allows JavaScript to respond to user actions such as clicks, enabling interactive and dynamic webpages.

5. ALGORITHM / PROCEDURE

1. Create an HTML file with elements to be manipulated: a heading, buttons and a text input.
2. Add a `<script>` block for the JavaScript logic.
3. Use `document.getElementById()` to select the elements to be modified.
4. Define a function to change the heading text dynamically.

5. Define a function to toggle the page background color.
6. Define a function to add new list items dynamically using `document.createElement()` and `appendChild()`.
7. Attach the functions to buttons using the `onclick` attribute.
8. Save the file and open it in a web browser.
9. Test the dynamic behavior by clicking each button.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Dynamic Webpage using DOM</title>
</head>
<body>
  <h1 id="heading">Welcome!</h1>
  <button onclick="changeText()">Change Text</button>
  <button onclick="changeColor()">Change Background</button>

  <h3>To-Do List</h3>
  <input type="text" id="taskInput" placeholder="Enter a task">
  <button onclick="addTask()">Add Task</button>
  <ul id="taskList"></ul>

  <script>
    function changeText() {
      document.getElementById("heading").innerHTML = "Text Changed Dynamically!";
    }

    function changeColor() {
      document.body.style.backgroundColor =
        document.body.style.backgroundColor === "lightyellow" ? "white" :
        "lightyellow";
    }

    function addTask() {
      const input = document.getElementById("taskInput");
      if (input.value.trim() !== "") {
        const li = document.createElement("li");
        li.textContent = input.value;
        document.getElementById("taskList").appendChild(li);
        input.value = "";
      }
    }
  </script>
</body>
</html>
```

7. CODE EXPLANATION

Tag / Code	Explanation
<code>document.getElementById()</code>	Selects an HTML element by its id attribute

innerHTML	Changes the HTML content inside an element
onclick	Attaches a function to run when the element is clicked
document.body.style	Accesses and modifies inline CSS styles via JavaScript
document.createElement()	Creates a new HTML element dynamically
appendChild()	Adds a newly created element as a child of another element
input.value	Reads or sets the value entered in a text input field

8. TEST CASES AND EXECUTION DATA

The webpage was opened in a browser and each interactive feature was tested by clicking the buttons and entering input.

Test Case	Description	Expected Result	Actual Result	Status
TC-01	Click "Change Text" button	Heading text changes	Heading text changed	Pass
TC-02	Click "Change Background" button	Background color toggles	Background color toggled	Pass
TC-03	Enter a task and click Add Task	Task appears in the list	Task appeared in the list	Pass
TC-04	Click Add Task with empty input	No empty item is added	No empty item added	Pass
TC-05	Add multiple tasks	All tasks are listed in order	All tasks listed in order	Pass

Test Case Output

TC-01 — *Heading text changes*

Text Changed Dynamically!

Change Text

Change Background

To-Do List

Enter a task

Add Task

Heading changed to 'Text Changed Dynamically!' after clicking the button

TC-02 — *Background color toggles*

Text Changed Dynamically!

Change Text

Change Background

To-Do List

Enter a task

Add Task

Background toggled to light yellow after clicking Change Background

TC-03 — *Task appears in the list*

Text Changed Dynamically!

Change Text

Change Background

To-Do List

Enter a task

Add Task

- Complete DOM assignment

'Complete DOM assignment' added to the to-do list

TC-04 — *No empty item is added*

Text Changed Dynamically!

Change Text

Change Background

To-Do List

Enter a task

Add Task

Clicking Add Task with an empty input added nothing to the list

TC-05 — All tasks are listed in order

Text Changed Dynamically!

Change Text

Change Background

To-Do List

Enter a task

Add Task

- Complete DOM assignment
- Revise JavaScript basics

Both tasks listed in the order they were added

9. OUTPUT

Output: Webpage Updating Dynamically

Welcome!

To-Do List

Initial page before any button is clicked

10. RESULT

Thus, a dynamic webpage was successfully developed using DOM manipulation and JavaScript, and the page content was verified to update dynamically without reloading.